



Takneek PS - Mid Prep



InfraPulse

Introduction:

Buildings, hostels, hospitals, and commercial complexes accumulate maintenance defects such as spalling, cracked tiles that often go unreported until they escalate into safety hazards or costly repairs. Facility teams currently rely on manual inspection and complaint reporting, which is slow, unstructured, and treats every complaint with the same urgency regardless of actual risk.

Problem Statement:

Participants must build a **Photo-Based Defect Detection & Priority Maintenance Web System** that will:

1. Allow a user to register a complaint with their name, address/location, a short problem description, and photograph of the defect.
2. The defect-detection model must analyse the submitted photograph to identify the visible defect type, map it to one of the three required categories, and route it to the designated staff queue.
 - **Structural:** spalling
 - **Functional:** stagnant water
 - **Performance:** paint peeling, cracked tiles (cracked tiles > paint peeling in priority)
3. Route the complaint to the correct category's dedicated staff queue.
4. Rank complaints within each category's queue by priority (e.g. visible severity, extent of the visible defect), and update the queue live as new complaints arrive.
5. Provide separate login portals for users and staff. Users can upload complaints (photo, name, address, description) and track their status; staff (logged into their respective domain Structural, Functional, or Performance) can view and manage only their category's live priority queue.
6. Let both the user (complaint status/position) and staff (full queue for their category) view current queue standing.

Deliverables:

The final solution is expected to provide the following:

1. **Complaint Registration Website**
 - Accepts user name, address/location, problem description, and photograph of complaint.
 - Separate login pages for User and Staff.



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- Staff must be able to update each complaint's status through predefined states — *Submitted* → *Assigned* → *In Progress* → *Resolved*. Once marked Resolved, the complaint is automatically removed from the live queue, and the user can view this final status on their end.
 - The system must automatically display the detected defect name and its corresponding category on the User and Staff portals.
2. **Defect Detection & Classification**
- Correct identification of visible defects and classification into Structural, Functional, or Performance.
 - Classification limited strictly to what is visibly evident in the photograph, no claims about non-visible or predicted defects.
 - Upon submission of a complaint photograph, the system must automatically identify and display the detected defect name (e.g., *Spalling*) and its corresponding category (e.g., *Structural*) on both the User and Staff portals. The displayed label and category must be generated by the system and must not require manual selection by the user or staff.
3. **Priority Queue System**
- Separate live queue per category (Structural, Functional, Performance).
 - Automatic ranking within each queue by defect priority (visible severity, visible extent).
 - Real-time updates as new complaints are submitted.
 - Queue visibility for users and staff.
4. **Documentation Report**
- Approach, detection logic, priority-ranking method, evaluation results, limitations, and suggestions for improving accuracy.

Submission Format:

Teams must submit:

- All source code, scripts, model weights/checkpoints, dependency files (requirements.txt / package.json), and setup instructions (README.md) involved in the detection/classification logic, web application (frontend + backend), and priority queue implementation must be submitted in a self-contained github repository and zip folder.
- A documentation and presentation detailing approach, challenges, and solutions in the same zip file named [PoolName_Takneek'26].
- Teams must provide a working URL of their hosted web application as part of the submission. The link must remain accessible throughout the evaluation period and must allow evaluators to access and test the complete system.



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Note:

- The complete system must be functional and accessible during evaluation. External AI/ML inference services or APIs that directly perform defect detection or classification are not permitted.
- General-purpose APIs and libraries may be used for supporting functionality such as authentication, databases, hosting, and other non-AI services, subject to proper citation/documentation.
- Any modification to the submitted repository, zip file, or hosted application after a valid on-time submission has been received will incur a 25% penalty on the team's final score.
- Any required submission file that is not submitted will result in 10% penalty on the team's final score.
- The Detection, Classification, Priority, and Queue functionality will be evaluated through the submitted web application.
- Any submission received after the deadline, even by one second, will incur a 10% penalty for the first minute or part thereof, with an additional 10% penalty for each subsequent minute or part thereof. Submissions received more than 10 minutes after the deadline will not be accepted.
- The evaluation dataset will be prepared by the leaders and will be used to test all submitted systems during evaluation.

Evaluation Metrics:

1. Detection & Classification (30%)

- The submitted web application must correctly identify the visible defect in the evaluation photograph (10%)
- The submitted web application must correctly classify the visible defect as Structural, Functional, or Performance (10%)
- The complaint must be automatically routed to the appropriate category's staff queue (10%)

2. Priority Queue Logic (40%)

- Correctly determines the relative priority of complaints based on the team's documented methodology (15%)
- Complaints appear in the appropriate order within the relevant staff queue according to the implemented priority logic (15%)
- Newly submitted complaints are automatically incorporated into the appropriate queue, without manual re-entry (10%)

3. Website Usability and Integration (20%)

- Registering and accessing a complaint (5%)
- Integration between user and staff portals (5%)
- Status tracking (5%)



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- UI/UX Design (5%)

4. Documentation (10%)

- Approach (2%)
- Detection logic (2%)
- Priority-ranking method (2%)
- Evaluation results (2%)
- Limitations (1%)
- Suggestions for improving accuracy (1%)

Rules and Team Composition:

1. **Team Size:** Maximum of 5 participants per team, with at least 2 Y26 and 2 Y25.
2. **Originality and Work Ownership:** All submissions must be original work. Use of external datasets and libraries must be clearly cited in the documentation. Teams may use generic, publicly available models or backbones pre-trained on general-purpose datasets (e.g., ImageNet, COCO) as a starting point. Use of any model or checkpoint pre-trained specifically on defect, stagnant water or building-damage datasets is not allowed and will lead to disqualification. Plagiarism or direct copying will lead to disqualification.
3. **Fair Play and Ethics:** Teams must not tamper with, manipulate, hardcode, memorize, or pre-associate evaluation inputs with predetermined outputs. The submitted solution must genuinely detect, classify, prioritize, and queue complaints from live inputs provided during evaluation.
4. **Tie-Breaker:** In case of a tie, priority will be given to the team with the higher score in Defect Detection & Classification, followed by Priority Queue Logic, and then Website Usability and Integration. If the tie persists, the final decision will be made by the leaders based on the overall demonstrated performance of the submitted system.

Submission Deadline:

1. **Deadline:** 3rd September, 11:59:59pm IST.
2. **Github Privacy:** Github repository must be private until deadline and made public after deadline.
3. **Demonstration and Presentation:** 4th September, 8pm IST.

Note:

For any queries, pool captains and PS leads are encouraged to ask in the communication channel.